Experiment 2: Generation of Square Wave and Sine Wave using Proteus

Aim:

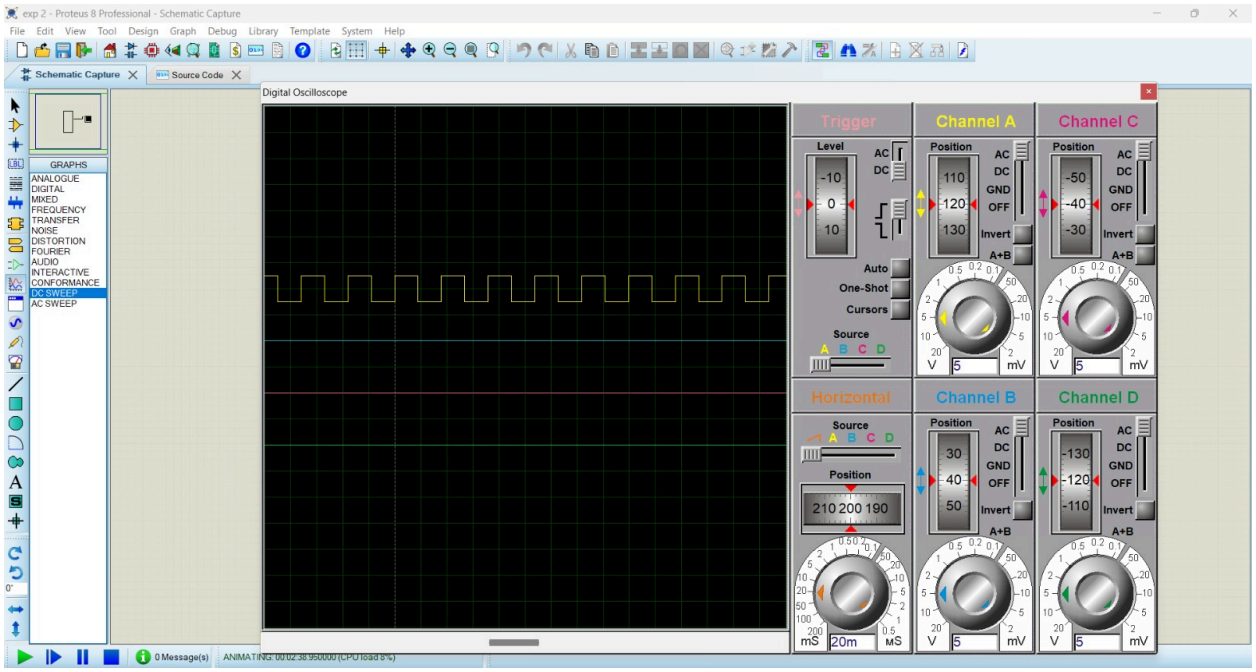
To generate a square wave using a microcontroller in Proteus.

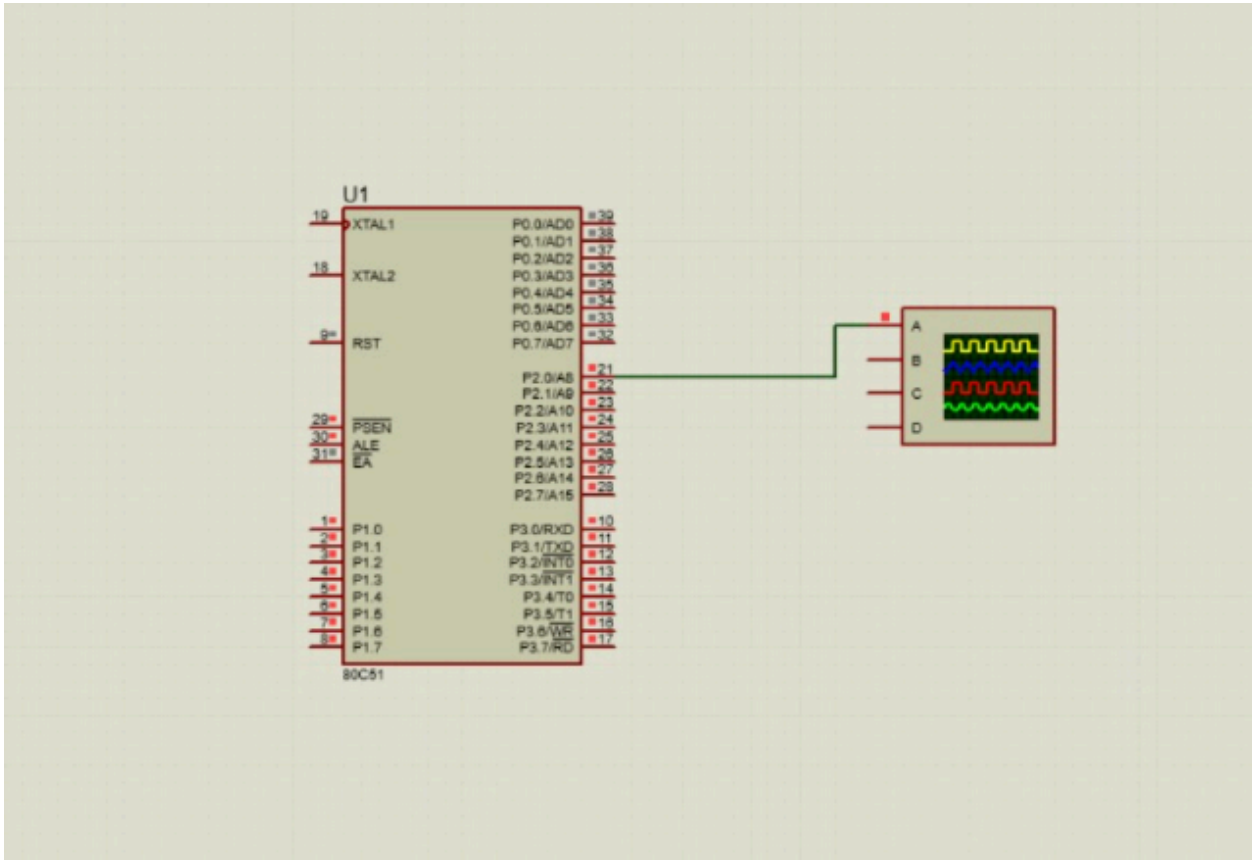
Procedure:

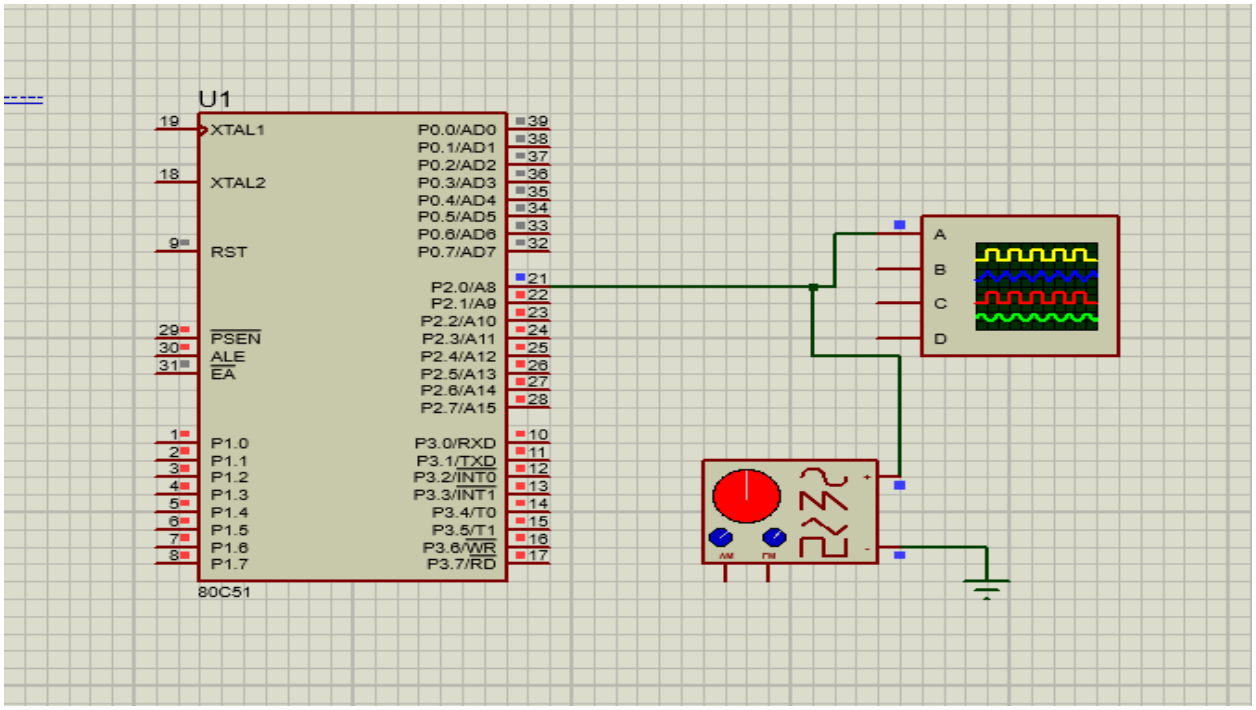
1. Open Proteus and create a new circuit.
2. Place AT89C51 and required components.
3. Connect an output pin to an oscilloscope.
4. Write and compile the square-wave generation program.
5. Load the HEX file into AT89C51.
6. Run the simulation and observe the waveform.

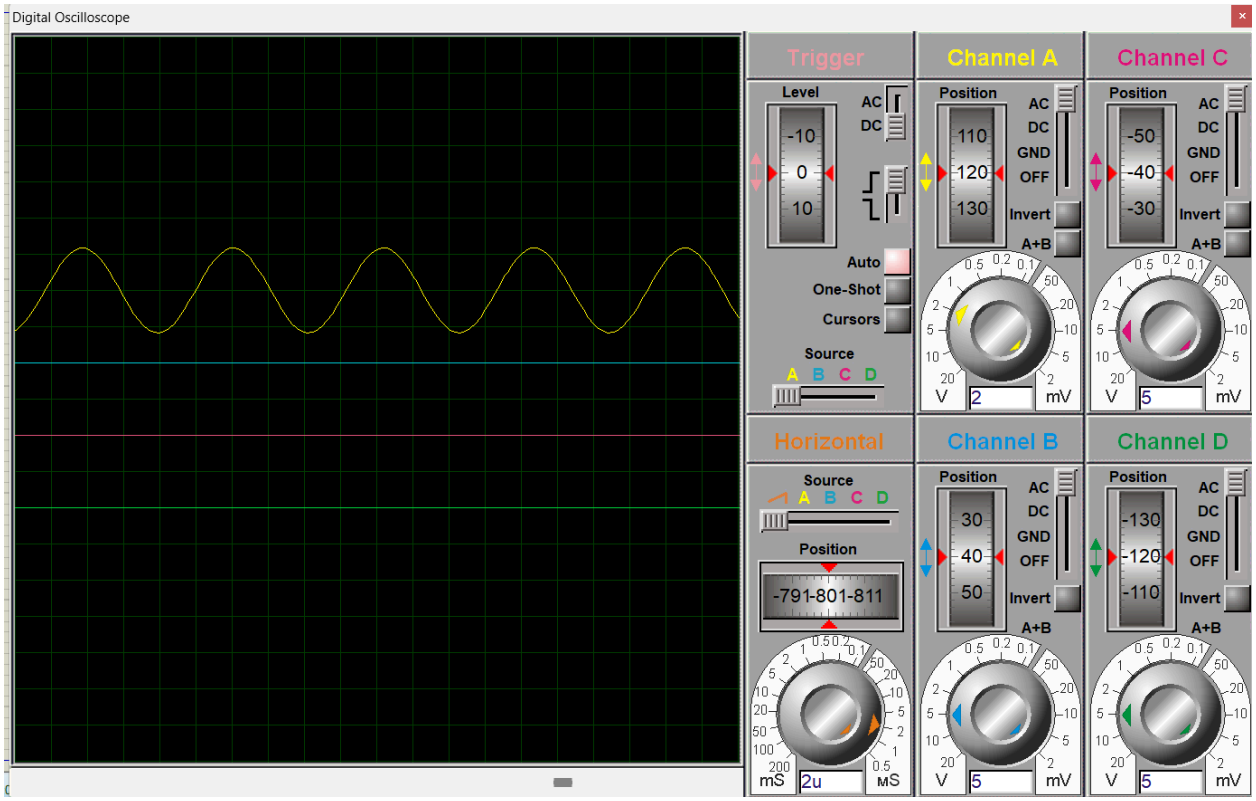
*main.asm

```
1   ORG 0000H
2
3   UP:     SETB P2.0
4           ACALL DELAY
5           CLR P2.0
6           ACALL DELAY
7           SJMP UP
8
9   DELAY:  MOV R4, #35
10
11  H1:     MOV R3, #255
12  H2:     DJNZ R3, H2
13          DJNZ R4, H1
14          RET
15
16  END
```


exp 2 - Proteus 8 Professional - Schematic Capture
File Edit View Tool Design Graph Debug Library Template System Help
Schematic Capture
Source Code
GRAPHS
ANALOGUE
DIGITAL
MIXED
FREQUENCY
TRANSFER
NOISE
DISTORTION
FOURIER
AUDIO
INTERACTIVE
CONFORMANCE
DC SWEEP
AC SWEEP
Digital Oscilloscope
Trigger
Level
Auto
One-Shot
Cursors
Source
Channel A
Channel C
Channel B
Channel D
Position
AC
DC
GND
OFF
Invert
A+B
Horizontal
210 200 190
20m
0 Message(s)


U1
XTAL1
XTAL2
RST
PSEN
ALE
EA
P1.0
P1.1
P1.2
P1.3
P1.4
P1.5
P1.6
P1.7
P0.0/AD0
P0.1/AD1
P0.2/AD2
P0.3/AD3
P0.4/AD4
P0.5/AD5
P0.6/AD6
P0.7/AD7
P2.0/A8
P2.1/A9
P2.2/A10
P2.3/A11
P2.4/A12
P2.5/A13
P2.6/A14
P2.7/A15
P3.0/RXD
P3.1/TXD
P3.2/INT0
P3.3/INT1
P3.4/T0
P3.5/T1
P3.6/WR
P3.7/RD
80C51
A
B
C
D





Result:

A square wave was successfully generated and observed on the oscilloscope.